
Булеан — множество $\mathcal{B} = \{0, 1\}$.

Булева функция — является отображением булевого пространства с размерностью n и количеством переменных n в аргументе функции

$$f : \mathcal{B}^n \rightarrow \mathcal{B} \quad f(x_1, x_2, \dots, x_n)$$

Сигма нотация булеана — определяется как

$$x^\sigma = \begin{cases} \bar{x}, & \text{if } \sigma = 0 \\ x, & \text{if } \sigma = 1 \end{cases}$$

СДНФ — Совершенная Дизъюнктивная Нормальная Форма, задаётся уравнением

$$f(x_1, x_2, \dots, x_n) = \bigvee_{(\sigma_1, \sigma_2, \dots, \sigma_n)} x_1^{\sigma_1} \wedge x_2^{\sigma_2} \wedge \dots \wedge x_n^{\sigma_n} \wedge f(\sigma_1, \sigma_2, \dots, \sigma_n)$$

СКНФ — Совершенная Конъюнктивная Нормальная Форма, задаётся уравнением

$$f(x_1, x_2, \dots, x_n) = \bigwedge_{(\sigma_1, \sigma_2, \dots, \sigma_n)} x_1^{\bar{\sigma}_1} \vee x_2^{\bar{\sigma}_2} \vee \dots \vee x_n^{\bar{\sigma}_n} \vee f(\sigma_1, \sigma_2, \dots, \sigma_n)$$

1. Найти СДНФ и СКНФ для $f_1(x, y, z) = x \implies (y \vee \bar{z})$

1.1. СДНФ для f_1

x	y	z	$\bar{z}$	$y \vee \bar{z}$	$x \implies (y \vee \bar{z})$
0	0	0	1	1	1
0	0	1	0	0	1
0	1	0	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

1. $\sigma = \{0, 0, 0\}, f_1(\sigma) = 1 \implies \overline{xyz} \cdot 1 = \overline{xyz}$

2. $\sigma = \{0, 0, 1\}, f_1(\sigma) = 1 \implies \overline{xyz} \cdot 1 = \overline{xyz}$

3. $\sigma = \{0, 1, 0\}, f_1(\sigma) = 1 \implies \overline{xyz} \cdot 1 = \overline{xyz}$

4. $\sigma = \{0, 1, 1\}, f_1(\sigma) = 1 \implies \overline{xyz} \cdot 1 = \overline{xyz}$

5. $\sigma = \{1, 0, 0\}, f_1(\sigma) = 1 \implies x\bar{y}\bar{z} \cdot 1 = x\bar{y}\bar{z}$

6. $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \implies x\bar{y}z \cdot 0 = 0$

7. $\sigma = \{1, 1, 0\}, f_1(\sigma) = 1 \implies xy\bar{z} \cdot 1 = xy\bar{z}$

8. $\sigma = \{1, 1, 1\}, f_1(\sigma) = 1 \implies xyz \cdot 1 = xyz$

Ответ: $f(x, y, z) = \overline{xyz} \vee \overline{xy}z \vee \overline{xy}\bar{z} \vee \overline{x}yz \vee x\bar{y}\bar{z} \vee xy\bar{z} \vee xyz$.

1.2. СКНФ для f_1

x	y	z	$\bar{z}$	$y \vee \bar{z}$	$x \implies (y \vee \bar{z})$
0	0	0	1	1	1
0	0	1	0	0	1
0	1	0	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

Ответ: $f(x, y, z) = \bar{x} \vee y \vee \bar{z}$.

1. $\sigma = \{0, 0, 0\}, f_1(\sigma) = 1 \implies x \vee y \vee z \vee 1 = 1$
2. $\sigma = \{0, 0, 1\}, f_1(\sigma) = 1 \implies x \vee y \vee \bar{z} \vee 1 = 1$
3. $\sigma = \{0, 1, 0\}, f_1(\sigma) = 1 \implies x \vee \bar{y} \vee z \vee 1 = 1$
4. $\sigma = \{0, 1, 1\}, f_1(\sigma) = 1 \implies x \vee \bar{y} \vee \bar{z} \vee 1 = 1$
5. $\sigma = \{1, 0, 0\}, f_1(\sigma) = 1 \implies \bar{x} \vee y \vee z \vee 1 = 1$
6. $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \implies \bar{x} \vee y \vee \bar{z} \vee 0 = \bar{x} \vee y \vee \bar{z}$
7. $\sigma = \{1, 1, 0\}, f_1(\sigma) = 1 \implies \bar{x} \vee \bar{y} \vee z \vee 1 = 1$
8. $\sigma = \{1, 1, 1\}, f_1(\sigma) = 1 \implies \bar{x} \vee \bar{y} \vee \bar{z} \vee 1 = 1$

1.3. Мн. Жегалкина для f_1 I способом

x	y	z	$\bar{z}$	$y \vee \bar{z}$	$x \implies (y \vee \bar{z})$
0	0	0	1	1	1
0	0	1	0	0	1
0	1	0	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

Ответ: $f(x, y, z) = 1 + xz + xyz$.

$$f(x, y, z) = a_0 + a_1x + a_2y + a_3z + a_{12}xy + a_{13}xz + a_{23}yz + a_{123}xyz$$

$$\begin{cases} a_0 = 1 \\ a_0 + a_3 = 1 \\ a_0 + a_2 = 1 \\ a_0 + a_2 + a_3 + a_{23} = 1 \\ a_0 + a_1 = 1 \\ a_0 + a_1 + a_3 + a_{13} = 0 \\ a_0 + a_1 + a_2 + a_{12} = 1 \\ a_0 + a_1 + a_2 + a_3 + a_{12} + a_{13} + a_{23} + a_{123} = 1 \end{cases} \begin{cases} a_0 = 1 \\ a_3 = 0 \\ a_2 = 0 \\ a_{23} = 0 \\ a_1 = 0 \\ a_{13} = 1 \\ a_{12} = 0 \\ a_{123} = 1 \end{cases}$$

1.4. Мн. Жегалкина II способом

$$f(x, y, z) = x \implies (y \vee \bar{z}) = \bar{x} \vee (y \vee \bar{z}) = \bar{x} \vee \bar{y}z = \overline{xy\bar{z}} = xy(z + 1) + 1 = 1 + xz + xyz$$

Ответ: $f(x, y, z) = 1 + xz + xyz$.

2. Найти СДНФ и СКНФ для $f_2(x, y, z) = (x \implies y) \vee (y \implies z)$

2.1. СДНФ для f_2

x	y	z	$x \implies y$	$y \implies z$	$(x \implies y) \vee (y \implies z)$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

Ответ: $f(x, y, z) = \overline{x}y\overline{z} \vee \overline{x}yz \vee \overline{x}\overline{y}z \vee \overline{x}yz \vee x\overline{y}\overline{z} \vee x\overline{y}z \vee xy\overline{z} \vee xyz$.

1. $\sigma = \{0, 0, 0\}, f_1(\sigma) = 1 \implies \overline{x}y\overline{z} \cdot 1 = \overline{x}y\overline{z}$
2. $\sigma = \{0, 0, 1\}, f_1(\sigma) = 1 \implies \overline{x}yz \cdot 1 = \overline{x}yz$
3. $\sigma = \{0, 1, 0\}, f_1(\sigma) = 1 \implies \overline{x}\overline{y}z \cdot 1 = \overline{x}\overline{y}z$
4. $\sigma = \{0, 1, 1\}, f_1(\sigma) = 1 \implies \overline{x}yz \cdot 1 = \overline{x}yz$
5. $\sigma = \{1, 0, 0\}, f_1(\sigma) = 1 \implies x\overline{y}\overline{z} \cdot 1 = x\overline{y}\overline{z}$
6. $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \implies x\overline{y}z \cdot 1 = x\overline{y}z$
7. $\sigma = \{1, 1, 0\}, f_1(\sigma) = 1 \implies xy\overline{z} \cdot 1 = xy\overline{z}$
8. $\sigma = \{1, 1, 1\}, f_1(\sigma) = 1 \implies xyz \cdot 1 = xyz$

2.2. СКНФ для f_2

x	y	z	$x \implies y$	$y \implies z$	$(x \implies y) \vee (y \implies z)$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

Ответ: $f(x, y, z) = 1$.

1. $\sigma = \{0, 0, 0\}, f_1(\sigma) = 1 \implies x \vee y \vee z \vee 1 = 1$
2. $\sigma = \{0, 0, 1\}, f_1(\sigma) = 1 \implies x \vee y \vee \overline{z} \vee 1 = 1$
3. $\sigma = \{0, 1, 0\}, f_1(\sigma) = 1 \implies x \vee \overline{y} \vee z \vee 1 = 1$
4. $\sigma = \{0, 1, 1\}, f_1(\sigma) = 1 \implies x \vee \overline{y} \vee \overline{z} \vee 1 = 1$
5. $\sigma = \{1, 0, 0\}, f_1(\sigma) = 1 \implies \overline{x} \vee y \vee z \vee 1 = 1$
6. $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \implies \overline{x} \vee y \vee \overline{z} \vee 1 = 1$
7. $\sigma = \{1, 1, 0\}, f_1(\sigma) = 1 \implies \overline{x} \vee \overline{y} \vee z \vee 1 = 1$
8. $\sigma = \{1, 1, 1\}, f_1(\sigma) = 1 \implies \overline{x} \vee \overline{y} \vee \overline{z} \vee 1 = 1$

2.3. Мн. Жегалкина для f_2 I способом

x	y	z	$x \Rightarrow y$	$y \Rightarrow z$	$(x \Rightarrow y) \vee (y \Rightarrow z)$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

Ответ: $f(x, y, z) = 1$.

$$f(x, y, z) = a_0 + a_1x + a_2y + a_3z + a_{12}xy + a_{13}xz + a_{23}yz + a_{123}xyz$$

$$\begin{cases} a_0 = 1 \\ a_0 + a_3 = 1 \\ a_0 + a_2 = 1 \\ a_0 + a_2 + a_3 + a_{23} = 1 \\ a_0 + a_1 = 1 \\ a_0 + a_1 + a_3 + a_{13} = 1 \\ a_0 + a_1 + a_2 + a_{12} = 1 \\ a_0 + a_1 + a_2 + a_3 + a_{12} + a_{13} + a_{23} + a_{123} = 1 \end{cases} \begin{cases} a_0 = 1 \\ a_3 = 0 \\ a_2 = 0 \\ a_{23} = 0 \\ a_1 = 0 \\ a_{13} = 0 \\ a_{12} = 0 \\ a_{123} = 0 \end{cases}$$

2.4. Мн. Жегалкина для f_2 II способом

$$f(x, y, z) = (x \Rightarrow y) \vee (y \Rightarrow z) = (\bar{x} \vee y) \vee (\bar{y} \vee z) = \bar{x}\bar{y} \vee \bar{y}z = \bar{x}\bar{y}z = \bar{0} = 1$$

Ответ: $f(x, y, z) = 1$.

3. Найти СДНФ и СКНФ для $f_3(x, y, z) = (x \wedge y) \Leftrightarrow (x \Rightarrow \bar{z})$

3.1. СДНФ для f_3

x	y	z	$\bar{z}$	$x \wedge y$	$y \Rightarrow \bar{z}$	$(x \wedge y) \Leftrightarrow (y \Rightarrow \bar{z})$
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	1	1
1	1	1	0	1	0	0

Ответ: $f(x, y, z) = \bar{x}yz \vee xy\bar{z}$.

- $\sigma = \{0, 0, 0\}, f_1(\sigma) = 0 \Rightarrow \bar{x}y\bar{z} \cdot 0 = 0$
- $\sigma = \{0, 0, 1\}, f_1(\sigma) = 0 \Rightarrow \bar{x}yz \cdot 0 = 0$
- $\sigma = \{0, 1, 0\}, f_1(\sigma) = 1 \Rightarrow \bar{x}y\bar{z} \cdot 0 = 0$
- $\sigma = \{0, 1, 1\}, f_1(\sigma) = 0 \Rightarrow \bar{x}yz \cdot 1 = \bar{x}yz$
- $\sigma = \{1, 0, 0\}, f_1(\sigma) = 0 \Rightarrow xy\bar{z} \cdot 0 = 0$
- $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \Rightarrow xy\bar{z} \cdot 0 = 0$
- $\sigma = \{1, 1, 0\}, f_1(\sigma) = 0 \Rightarrow xyz \cdot 1 = xyz$
- $\sigma = \{1, 1, 1\}, f_1(\sigma) = 1 \Rightarrow xyz \cdot 0 = 0$

3.2. СКНФ для f_3

x	y	z	$\bar{z}$	$x \wedge y$	$y \implies \bar{z}$	$(x \wedge y) \iff (y \implies \bar{z})$
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	1	1
1	1	1	0	1	0	0

Ответ: $f(x, y, z) = x \vee y \vee z \wedge x \vee y \vee \bar{z} \wedge x \vee \bar{y} \vee z \wedge \bar{x} \vee y \vee z \wedge \bar{x} \vee y \vee \bar{z} \wedge \bar{x} \vee \bar{y} \vee \bar{z}$.

1. $\sigma = \{0, 0, 0\}, f_1(\sigma) = 0 \implies x \vee y \vee z \vee 0 = x \vee y \vee z$
2. $\sigma = \{0, 0, 1\}, f_1(\sigma) = 0 \implies x \vee y \vee \bar{z} \vee 0 = x \vee y \vee \bar{z}$
3. $\sigma = \{0, 1, 0\}, f_1(\sigma) = 0 \implies x \vee \bar{y} \vee z \vee 0 = x \vee \bar{y} \vee z$
4. $\sigma = \{0, 1, 1\}, f_1(\sigma) = 1 \implies x \vee \bar{y} \vee \bar{z} \vee 1 = 1$
5. $\sigma = \{1, 0, 0\}, f_1(\sigma) = 0 \implies \bar{x} \vee y \vee z \vee 0 = \bar{x} \vee y \vee z$
6. $\sigma = \{1, 0, 1\}, f_1(\sigma) = 0 \implies \bar{x} \vee y \vee \bar{z} \vee 0 = \bar{x} \vee y \vee \bar{z}$
7. $\sigma = \{1, 1, 0\}, f_1(\sigma) = 1 \implies \bar{x} \vee \bar{y} \vee z \vee 1 = 1$
8. $\sigma = \{1, 1, 1\}, f_1(\sigma) = 0 \implies \bar{x} \vee \bar{y} \vee \bar{z} \vee 0 = \bar{x} \vee \bar{y} \vee \bar{z}$

3.3. Мн. Жегалкина для f_3 I способом

x	y	z	$\bar{z}$	$x \wedge y$	$y \implies \bar{z}$	$(x \wedge y) \iff (y \implies \bar{z})$
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	1	1
1	1	1	0	1	0	0

Ответ: $f(x, y, z) = xy + yz$.

$$f(x, y, z) = a_0 + a_1x + a_2y + a_3z + a_{12}xy + a_{13}xz + a_{23}yz + a_{123}xyz$$

$$\begin{cases} a_0 = 0 \\ a_0 + a_3 = 0 \\ a_0 + a_2 = 0 \\ a_0 + a_2 + a_3 + a_{23} = 1 \\ a_0 + a_1 = 0 \\ a_0 + a_1 + a_3 + a_{13} = 0 \\ a_0 + a_1 + a_2 + a_{12} = 1 \\ a_0 + a_1 + a_2 + a_3 + a_{12} + a_{13} + a_{23} + a_{123} = 0 \end{cases} \begin{cases} a_0 = 0 \\ a_3 = 0 \\ a_2 = 0 \\ a_{23} = 1 \\ a_1 = 0 \\ a_{13} = 0 \\ a_{12} = 1 \\ a_{123} = 0 \end{cases}$$

3.4. Мн. Жегалкина для f_3 II способом

$$\begin{aligned} f(x, y, z) &= (x \wedge y) \iff (y \implies \bar{z}) = xy \iff \bar{y}\bar{z} = (xy \implies \bar{y}\bar{z})(xy \iff \bar{y}\bar{z}) = \overline{xy\bar{y}\bar{z}} \cdot \overline{\bar{x}\bar{y}} \cdot \overline{\bar{y}\bar{z}} = \\ &= (xyz + 1)((xy + 1)(yz + 1) + 1) = xyz(xy + 1)(yz + 1) + xyz + (xy + 1)(yz + 1) + 1 = \\ &= xyz(xyz + xy + yz + 1) + xyz + xyz + xy + yz + 1 + 1 = \\ &= xyz + xyz + xyz + xyz + xy + yz = xy + yz \end{aligned}$$

Ответ: $f(x, y, z) = xy + yz$.